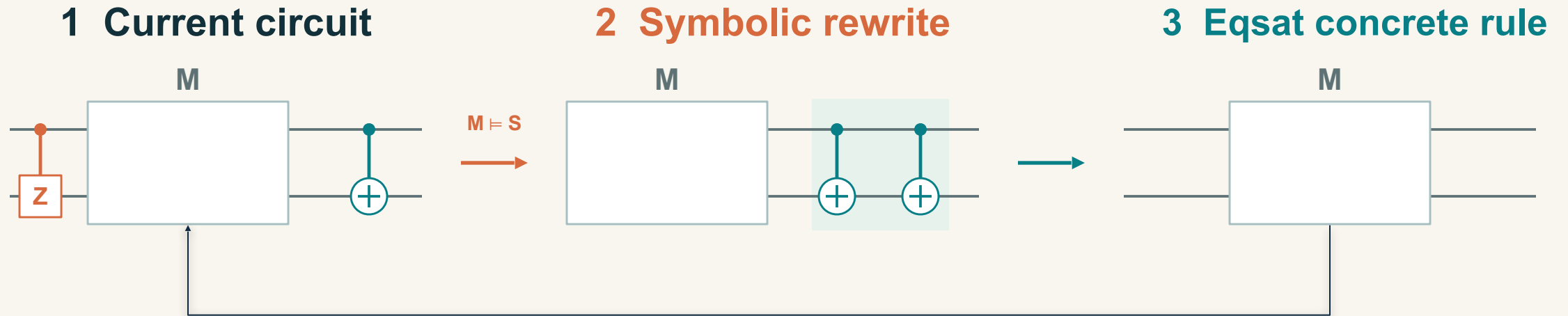


Simple Optimization Algorithm

One search step: rearrange, then simplify



Symbolic Rewrite move/transform gates from one place to another;
Concrete rules extract locally optimal by equality saturation

Evaluation setup

135

benchmark circuits

4–36

qubits per circuit

60 min

search budget · 3 trials

Benchmarks and targets

QAOA, VQE, QPE, QFT,
Grover and Shor families

IBM-Eagle and Nam

15–17,438 input gates

QSymb search

Up to 9 equality-saturation iterations,
alternating with symbolic rewrites